

Business Case: Cutting Post-Harvest Losses with Predictive Analytics

Fresh Supplies Platform | Prepared for: Kirinyaga Cooperative Board & CFO | Prepared by: Amos Ndungo | Week 11

PROBLEM STATEMENT

Kirinyaga Cooperative loses an estimated **25% of its harvested produce to spoilage** before it reaches a buyer, and much of what does sell is delivered to the nearest market rather than the best-paying one. On the cooperative's roughly 500-tonne annual harvest, this costs an estimated **KES 5,000,000+ every year** in produce that generates zero revenue, plus additional revenue left on the table from under-priced sales.

PROPOSED SOLUTION

A farmer logs a harvest — crop, quantity, and basic storage/transport conditions — using either a phone app or a simple text-message menu (no smartphone or internet required). Within seconds, the system estimates the batch's spoilage risk and tells the farmer, in one sentence, which nearby market will pay the most for it before it spoils. No change to how produce is grown, stored, or transported — only better information delivered at the moment the farmer decides where to sell.

FINANCIAL IMPACT

Assumptions (conservative, shown for transparency): 500,000 kg annual harvest volume; KES 40/kg farm-gate value; current spoilage rate 25%; solution reduces spoilage by 30% relative (to 17.5%) and lifts average realized price by 8% via better market matching.

	Sold Volume	Avg. Price/kg	Realized Revenue
Baseline (today)	375,000 kg	KES 40.00	KES 15,000,000
With Fresh Supplies	412,500 kg	KES 43.20	KES 17,820,000

Annual Financial Benefit = KES 17,820,000 – KES 15,000,000 = **KES 2,820,000/year**

Cost Item	Year 1	Year 2+
Platform subscription	KES 300,000	KES 300,000
Onboarding and training (one-time)	KES 150,000	—
Total Cost	KES 450,000	KES 300,000

Year-1 ROI	Payback Period	Year-2+ Net Annual Savings
526%	~1.9 months	KES 2,520,000/yr

RISKS & MITIGATION

Risk	Mitigation
1. Projections are based on a calibrated model, not yet field-validated on Kirinyaga's own harvests — real results in year one could be lower than forecast.	Run a 90-day pilot at one collection point first. Compare actual outcomes to the forecast before committing to cooperative-wide rollout.
2. Some farmers lack smartphones or reliable internet, which could limit adoption.	The system already works over a basic text-message menu — every farmer can participate on any phone, with no data plan required.

RECOMMENDATION

GO — phased. Approve a 90-day pilot at one collection point (low cost, payback in under 2 months even at pilot scale) before committing to the full KES 450,000 first-year, cooperative-wide rollout.